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RESEARCH ARTICLE

Alleviate: An AI-Powered Mobile Application for Personalized Healthcare Management and Doctor Recommendation

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ABSTRACT The areas of access and control in the healthcare industry are probably one of the most significant areas that could greatly benefit from advancements in technology. The mobile application developed in this paper is based on the latest research techniques aimed at providing comprehensive health care services to its users, including patients and healthcare providers. This application was built using Flutter and Dart for the frontend and Flask for the backend along with Firebase for data storage. It uses machine learning techniques like Logistic Regression (LR) and Random Forest (RF) to create predictive models for specialists, dietary habits and physical exercise using carefully curated datasets, enriched by the implementation of Natural Language Processing (NLP) techniques achieving 97.24% precision using Logistic Regression Model (LR). In addition to this, it also supports doctor authentication through secure web automation since it authenticates physicians through the Indian Medical Registry (IMR). The application has many functional features such as scheduling appointments, managing prescriptions, and using a clinic dashboard, which make it more useful for healthcare professionals. It also provides personalized diet and exercise plans based on the individual health metrics as well as location-based features, using OpenStreetMap and geospatial analysis to help navigate to the nearest healthcare services. It also facilitates a report generation system with artificial intelligence that equips health care professionals with an overall history of a patient's medical past, thus supporting both diagnosis and treatment plans. The range of features in this application like the ones listed above makes the solution one of a kind. The innovative aspects of the application like voice recognition technology and QR code scanning introduce various key enhancements in terms of current systems. This comprehensive healthcare solution utilizes advanced technology to bridge gaps in healthcare services, offering a secure and user-friendly platform for patients and healthcare professionals alike.

INDEX TERMS AI-ML, alleviate, BMI, clinical mobile application, EHR, firebase, fitness planner, flask, flutter, Groq API, IMR, instant appointment booking, LR, NLP, OSM, nearby pharmacy, personalized clinic finder, prescription management, QR code, RF, symptom based specialist finder.

I. INTRODUCTION

Delivery of healthcare still presents major challenges due to fragmented systems, and the advent of the digital era has created a tremendous surge in demand for user-friendly programs for augmenting medical procedures. There is a

clear need for sophisticated solutions enabling processes like finding appropriate healthcare providers, arranging same-day appointments, tracking prescriptions, and tracking individualized health records.

Alleviate is intended to improve access to healthcare through linking users with professional medical professionals, offering customized diet and exercise regimens, instant medical counseling based on user-reported symptoms,

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and facilitating integration of healthcare services in a seamless way.

In this backdrop of challenges, Alleviate proposes a mobile app that integrates sophisticated machine learning models with clinical features, real-time synchronization features, personalized clinic location features, and efficient management of electronic health records (EHR). The key features of Alleviate include symptom-based physician recommendations enabled by machine learning algorithms, voice input support for users to help people with various literacy and linguistic levels, personalized diet and exercise recommendations based on user health data (activity level, height, weight, and full health profile), and hospital, clinic, and pharmacy integration for comprehensive access to healthcare services. In addition to this, the app provides users with instant booking functionalities based on specialist schedules and offers navigation support through OpenStreetMap services, along with locating near pharmacies for easy access to drugs.

In addition, Alleviate has a wide-range prescription management feature. On their dashboard, users can download prescriptions from healthcare practitioners, while doctors can author prescriptions according to editable templates and access patient history. A registered user can manage prescriptions for different members of a family, thus encouraging overall healthcare management for the whole family. The platform also includes an AI-driven virtual assistant called Allevia, which engages in a dynamic, interactive conversation with users, providing answers to health questions and live recommendations.

II. RELATED WORKS

A. LITERATURE SURVEY

Tan et al., [1] developed a comprehensive medical portal that implements a waterfall model for the system development cycle which targets a large audience. Implementation involved using Flask, Python, HTML, CSS, JavaScript for the user interface and SQLite for data management. A machine learning model was integrated for disease diagnosis, personal health record management, and medicine reminders. They utilized openAI GPT 3.5 to enhance the portal providing drug interaction checks and informative user experience. Various preprocessing techniques were used on the data set to improve the accuracy of the model. Prognosis served as a label for a random forest algorithm that predicted disease based on symptoms provided by the user as input. They also used a GPT-powered chatbot which is capable of restoring all the chat history where the user can be engaged in natural language conversation. The future scope includes the enhancement of the navigation of hospitals through maps instead of text format.

Abinaya and Ganesan [2] proposed a method leveraging the R-tree algorithm for efficient data fetching and spatial filtering, ensuring unwanted locations are discarded before the search space. They focus on making an Android application that helps in locating the nearest healthcare

facility during an emergency by determining the shortest route from source to destination using k-nearest neighbor queries and Ahuja-Dijkstra's algorithm. Data retrieval is implemented with SQL, Python and JSON, while the Android app is developed using Eclipse and Apache Tomcat as the server, facilitating communication between the server and the mobile device. The database included special coordinate addresses and pin codes of the healthcare facilities; their future scope involves integrating Google API for optimal routing and connecting facilities for enhanced navigation.

Munir et al., [3] developed an Android application that aimed to reduce time and improve cost efficiency for patients by locating hospitals that are within a range of 5 km radius offering desired specialists. Using GPS and Google API services, the application determines the shortest route from the current location to the hospitals. Data regarding 18 major hospitals in Karachi is collected and the data includes hospital names, a list of specialized doctors, contact details, and addresses which are stored in a database. The app features a category screen with predefined specialists and their comprehensive profiles. Testing revealed that the specialized hospital was primarily located in the center of the city. Their future announcements included real-time doctor availability expanding beyond Karachi and introducing online appointments to save patients time.

Nimbalkar and Fadnavis [4] mainly focused on addressing medical emergencies by enabling users to locate the nearest hospitals and share their critical health information instantly with a selected hospital. The system activates clients' requests captures the user's coordinates and emergency type and uses A* algorithm to locate hospitals in the shortest possible route. All hospital data is stored and updated regularly in the cloud. Electronic health data is generated by the click of an emergency button which is sent to a selected hospital for pre-medical treatment. This gives an instant prescription from the doctor for timely treatment. User authentication is mainly based on role and these SQL queries are executed via PHP as a backend database. This application mainly emphasizes real-time EHR maintenance ensuring accurate updated health records. Their future announcement includes with efficient distance tracking improvised hospital information management clinic module and a more user-friendly interface.

Kumar et al., [5] This work uses machine learning models to predict diseases based on the symptoms given by users, and it recommends doctors based on various fields such as location, fees, etc. It successfully provided doctor recommendations and predicted diseases based on symptoms given. The work indicated that the Random Forest model gets the highest accuracy, and it has integrated classification algorithms to make user experience more friendly; on the other hand, the work was limited based on the unavailability of proper datasets, and a lot of the implemented models displayed low accuracy.

Björkdahl et al., [6] This work implemented the usage of small language models (SLMs) to process multimodal

inputs and predict disease risk. It juggles multiple tasks simultaneously and focuses on the fusion of multiple data modalities. The work successfully provided competitive performance when exposed to multiple modalities. Though multimodality is an exceptional feature of the model, it does not outperform state-of-the-art models tailored for a single task. The model is very scalable and handles multiple disease predictions; it handles unbalanced data well too. The model has very high computational costs, in part due to its multimodality.

Umbare et al., [7] This work incorporates AI and ML-powered systems for disease prediction, generation of prescription, and location based services. It uses tools such as NLP (Natural Language Processing) for symptom processing, KNN (K-Nearest Neighbours) for disease classification and risk assessment, the work also implements a Dialogflow Chatbot for real time interaction. The usage of NLP handles unstructured data efficiently and the presence of real time Chatbot enhances the user experience and healthcare access. The prescription generation feature adds a personalized touch to the experience. Although the work has limitations KNN struggles with high dimensionality data making scalability an issue and the need of specialized datasets are required for deep learning results.

Gujar et al., [8] The work applies a Naive Bayes classifier for disease prediction based on symptoms such as blood pressure, high blood sugar, etc. Weka is used for model training, and CoreNLP is used for doctor reviews and recommendations. The dataset used comprises features ranging from gender and blood pressure to pulse rate and symptoms such as fever and headache. This model is convenient for disease prediction and doctor predictions, hence saving time. Easily customizable doctor recommendations based on user reviews, Naive Bayes classifiers struggle with complex interrelated symptoms.

Mogaveera et al., [9] The work combines health monitoring with diet planning and exercise plans, it uses C4.5 decision tree algorithm to generate diet and exercise plans based on user inputs such as height, weight, and medical history. This also includes regular health monitoring and follow-ups for conditions like diabetes. The use of the C4.5 tree provides accurate and fast recommendations that would be tailored to the requirements of the user based on user history. The implementation of continuous follow-ups to track progress is a user-friendly touch. The scope of the model is very limited, though, and the dataset restricts flexibility.

Tanwar et al., [10] This paper highlights how Artificial Intelligence (AI) and Machine Learning (ML) can be applied in healthcare through chatbots. The chatbot was built using Google Dialogflow which provides a prompt bar and responds to health related queries. Before consultation with a doctor, it helps the user to deal with minor health problems such as colds, headaches etc using NLP. It also focused on the evolution of chatbots from ELIZA (an early system) to the current AL models. The chatbot emphasizes information

organizing, management of responses and smooth communication between user and agent. The paper also deals with the issues such as requirements for model training, restrictions and reply controls. Finally, it aims to increase accessibility and immediate medical advice.

Surve et al., [11] This study discovers the usage of ML algorithms such as KNN, SVM, and Decision Trees for prediction of diseases based on the symptoms given by the user. The system includes both text and voice input prompts using NLP techniques like tokenization and stemming. Algorithms such as Cosine similarity and TF-IDF are applied to improve the response accuracy. The Kaggle dataset used in this system has 4920 records of 41 diseases which includes symptom-disease mapping. The results based on KNN algorithm had 98.06 % of accuracy which is very efficient for disease diagnosis based on user symptoms. Although the system excels in the above features, the small dataset is its drawback. Further training is needed to enhance accuracy and handle more complex medical cases.

Mendapara et al., [12] This paper deals with the usage of NLP techniques to analyze user symptoms and provide disease diagnosis. ML models are employed to predict diseases and recommend the necessary treatments. The chatbot includes a conversational interface that guides the users through their health queries. The system utilizes live healthcare datasets, including real-time patient symptoms and diagnoses (Electronic Health Report) to incorporate and improve predictions. The chatbot lowers healthcare expenses, enhances accessibility, and offers round-the-clock support for immediate responses. It lacks advanced security measures like as biometrics and is only capable of text-based communications, with no voice capability.

Rahman et al., [13] developed a web-based disease prediction system by integrating deep learning techniques through transfer learning and conversational AI via Dialogflow to improve medical suggestions. The system draws relevant features for the prediction of diseases by using transfer learning from pre-trained deep models. The system incorporates Dialogflow in order to implement the interface of interaction with the user and allow users to submit symptoms conversationally, thus improving engagement and data collection. One of the keys to attaining great computational efficiency is transfer learning, and Dialogflow enables the creation of the conversational interface. A multi-disease dataset is used to test the system's performance, showing promise in terms of precise sickness prediction and tailored therapy recommendations. The method improves accessibility and streamlines the diagnostic process by bridging the gap between patients and healthcare providers. Although the method is novel, its effectiveness is highly dependent on the caliber of user input and pre-trained models. The conversational interface may encounter challenges when dealing with intricate medical terminology or a multitude of linguistic inputs. It would be more useful if diseases are further scoped and real-time data sources are included.

Tseng et al., [14] designed an interactive healthcare system that tailored diet and exercise advice suited to the individual needs of patients with chronic diseases, mainly focusing on metabolic syndrome. It supports real-time health monitoring as it collects key health indicators uploaded by the user. The system offers personalized advice on managing chronic conditions most effectively. The research included designing both a browser based platform and a mobile app, and hence ensure access and convenience in healthcare management. The system includes interactive modules for inputting and analyzing health data using algorithms to assess the health risks and delivery of targeted recommendations. Its mobile and web interfaces enable easy communication and interaction with users. It is validated by an Institutional Review Board for human subject research. The low cost, user-friendliness, and extensibility to various chronic diseases underpin its potential as a practical solution for personalized healthcare. The user compliance and adequacy of self-reported data will significantly determine the effectiveness of the system. Further, enlarging the model to suit other chronic conditions or incorporating sophisticated predictive analytics may not be an easy task.

Bhanuteja et al., [15] proposed a machine learning system for disease diagnosis based on the symptoms given for improvement in the early detection of diseases and decisions in healthcare. The analysis was done on a dataset with 4,920 patient records and 41 diseases, where they selected 95 highly significant symptoms out of 132 taken for the study. They utilized three machine learning algorithms, that are Random Forest, Decision Tree Classifier, and LightGBM, to build and evaluate the prediction model. The model successfully demonstrated precise disease prediction, thereby assisting healthcare professionals in early detection, better patient care, and decision-making. It also emphasized the reliability of machine learning in processing large-scale medical data. It fails to discuss issues like real-time implementation, scalability on higher datasets, or inclusion of dynamic real-world variables such as lifestyle and environmental factors that may affect diagnosis. Besides this, the report states it achieves efficiency, however, there is no overview of detailed metrics or comparative studies with similar models.

B. METHODOLOGY

As shown in Fig. 1; The mobile application is an integrated solution that uses Flutter and Dart for its front end, for seamless user experiences, and the establishment also includes connecting to several different components of the system to ensure that user functionality and interactions remain uninterrupted. The back end is built on the Python web framework Flask, which places reliance on machine learning algorithms for predictive modeling, the use of Logistic Regression (LR) for recommendations regarding specialists, and Random Forest (RF) for personal diet and exercise routines. All these get trained and finalized with a robust dataset stored within the system that enables accurate

data-driven predictions. The Flask backend also facilitates the use of verification of doctors by utilizing Web Automation techniques. To further enhance the backend, Backend as a Service (BAAS) platform, Firebase was integrated. Firebase is used for efficient data storage, user authentication, and synchronization of real-time data, and through this, allows the application to provide reliable and customized solutions to the needs of its users. The system also includes the API Services module, with OpenStreetMap (OSM) [16] which happens to be elementary in offering data on geospatial and external information to make the application even richer in functionality. API Services implementation ensures that location-based features and external data are well integrated within the user experience. This helps the user to navigate to the hospitals or clinics with a click of a button.

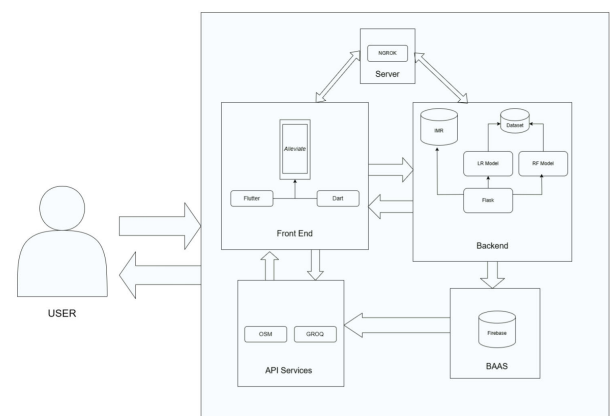


FIGURE 1. Application architecture.

1) USER VALIDATION

Our application consists mainly of users and doctors. For doctors, a robust validation process is implemented with the help of Web Automation to authenticate their credentials through the Indian Medical Registry (IMR) database [17]. It is a central database that keeps a record of all registered medical professionals in India. It is maintained by the National Medical Commission and ensures that only doctors with valid licenses can practice medicine. This process ensures that only valid professionals are registered on the platform and provides transparency in the healthcare system.

2) DATA PREPARATION AND MODEL TRAINING

The IEEE Data Port [18] provided this dataset. There are 1326 columns with symptoms in it, and the last column is titled "Prognosis." There are 415 rows of distinct prognoses in the dataset, with a prognosis for every symptom. Binary values are used to represent symptoms, with 1 denoting the presence of a symptom for a given disease and 0 denoting its absence.

We divided the dataset into 80% for training and 20% for testing, following normal procedures. Using this data, we first trained a Random Forest Classifier, which produced

a 62% accuracy rate. In a similar vein, we trained a 64% accurate logistic regression model. When these two models were combined, the accuracy was 63%. Because each prognosis was specifically mapped to a particular collection of symptoms, we conducted additional analysis and concluded that Logistic Regression was more appropriate for our dataset which uses multinomial classification to predict disease.

To increase the accuracy we realized that instead of using 1's and 0's, we could introduce a new column Symptom Mapping where the textual values of the symptoms were stored in a list for each prognosis. However, we got 70% accuracy after training for these two columns. Later we realized data augmentation was required so we used the `nlpaug` module to generate a list of synonyms of the symptoms mapping to each prognosis. For each prognosis, we introduced a new column called Specialists with the help of GROQ api. We covered all types of specialists mapped for that particular prognosis and got a list of specialists for each row.

By introducing hundreds of synonyms for each symptom, we significantly expanded the knowledge base. After training, we achieved nearly 97.24% precision using a logistic regression model which showed better prediction compared to the random forest classifier. Logistic Regression is used for classification tasks where the probability of an instance belonging to a certain class is modeled using the logistic (sigmoid) function. The mathematical representation is:

$$P(Y = 1 | \mathbf{X}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n)}}$$

where:

- $P(Y = 1 | \mathbf{X})$ is the probability of the class label being 1 given the input feature vector $\mathbf{X}$.
- β_0 is the intercept (bias term).
- $\beta_1, \beta_2, \dots, \beta_n$ are the learned coefficients corresponding to features $X_1, X_2, \dots, X_n$.
- e is Euler's number (approximately 2.718).

$$P(Y = k | \mathbf{X}) = \frac{e^{(\beta_k \cdot \mathbf{X})}}{\sum_{j=1}^K e^{(\beta_j \cdot \mathbf{X})}}$$

where k is the class label and K is the total number of possible classes (diseases in our case).

3) TRAINING: GOOGLE COLAB

This is a cloud-based platform, offering free GPU's and CPU's which are beneficial in training the model effectively, provides easy integration of google drive which allows for storing, accessing the dataset. The code and results can be shared between the team, ensuring the reproducibility of the project but there are several limitations such as session timeouts, memory restrictions.

4) LOGISTIC REGRESSION ARCHITECTURE

We performed a multi-class classification of medical data using Multinomial Logistic Regression model with

scikit-learn. The model works on a numerical feature matrix obtained from medical data such as test results and symptoms. Text data is transformed through vectorization. We compute logits for each disease and use the softmax function to convert logits into a probability distribution, and the class with the highest probability is considered the final prediction. The training utilizes the solver with log loss minimization, maximum 500 iterations, and class balancing (class weight=balanced) as an antidote for imbalanced datasets. This model, once trained, provides predictions for the disease classes for new samples and their respective confidence scores with respect to the classes predicted. Performance assessment measures, including precision, recall, f1 score, and support, allow for a rigorous assessment of performance, especially when data are imbalanced.

5) DOCTOR'S PROFILE

When a doctor registers on our app, they can manage their profile with various options. They can update their specialization, bio, address, phone number, and location, or choose to delete their profile if needed. The profile also includes two key statuses:

- Availability Status: When enabled, users can contact the doctor directly.
- Appointment Status: When enabled, users can book appointments by selecting available slots.

6) APPOINTMENTS SCHEDULING

In the doctor's dashboard, there is a section called Appointments that allows doctors to manage their availability for patient visits. In this section, the doctor can enter the total number of patients to be seen per day, which is the same as the total number of appointment slots. Doctors can select the clinic's opening and closing times.

Based on the selected timings, appointment slots are automatically generated with one-hour gaps. After the configuration, the doctor can save the settings by clicking the Save button. It also includes an Edit button, enabling doctors to update the number of patients or adjust the timings. These appointment slots are then reflected on the user side when they search for doctors and proceed to book appointments.

7) E-PRESCRIPTION

Our app includes a prescription feature that simplifies the process for doctors and patients. Doctors can create templates for prescriptions, making it easier to manage frequent cases. After diagnosing a disease or symptom, the doctor can write a prescription directly through the app by entering the patient's registered email ID and their name. They can describe the identified problem and add details about the prescribed medicines. The doctor can specify whether each medicine should be taken before or after meals using checkbox buttons. Once all the fields are filled, the prescription can be saved. After the doctor issues a prescription, the user can download and view the PDF directly from their dashboard, and also the user can save the PDF for future checkups. This feature

ensures that individual prescriptions can be created for each family member under the registered user, providing prescription management easy for every family member.

8) CLINIC MANAGEMENT

A doctor can add multiple clinics or organizations where they work. This can be done by filling a form that takes details regarding the clinic like its location and other information. This creates a separate dashboard for every clinic a doctor works in, making patient management more structured and can be accessed easily.

9) DIET AND EXERCISE PLAN

The user is provided a special Diet Plan page where they can define their activity level, following which a customized plan is recommended. It takes into account the health factors of the user, like their height, weight, age, and gender, from their account. A Random Forest model, which is trained on publicly available nutrition and fitness data [19], analyzes these factors to arrive at a prediction of the user's daily calorie needs with a precision of 84.76%, which in turn is utilized to create a three course meal plan. Presently, the meals are optimized to include South Indian cuisine, primarily for users in Bangalore and surrounding areas. Likewise, the Exercise Tips page advises users with different Body Mass Index (BMI) levels with suggested exercises to use for appropriate intensity levels and gives five customized exercise recommendations. Currently, the ML model is integrated with GROQ API, which uses the llama model to create a meal plan from the calories predicted by the model. Future work could be to build a custom model to create the meal plan instead of relying on an open source LLM.

10) AI BASED REPORT VIA QR SCAN

To quickly summarize the health records of a patient for the doctor, the patient can show the Quick Response (QR) code in their profile. When scanned, the QR code displays all the health metrics of the user along with an AI generated report of the patient's medical history. The QR code is generated using the barcode scanner package in Flutter. If the code is scanned anywhere outside the app, no information is leaked, so only the doctor has access to the health records, ensuring security.

To facilitate the assessment of the patient, an AI report is generated based on the prescriptions that other doctors have given to the patient in the past. This enables the doctor to gain deeper insights into the patient's medical history for better evaluation and care.

11) GEO SPATIAL POINTS

The application, through geolocator in conjunction with OSRM for routing and geospatial analysis, will be able to identify the health service providers within a defined kilometer range of the user's current location. Geo-points are calculated in real-time, and using OpenStreetMap, it ensures that the results are accurate and user-centric.

12) SELECT LOCATION DURING REGISTRATION

During user registration, the app uses `open_street_map_search` and `pick` to allow a user to pick a location on a map interface. This makes sure that the user can set his or her exact location, which is then saved to database known as `cloud_firestore` for future reference. The selected location is used to personalize the user's experience by suggesting relevant healthcare services nearby.

13) NAVIGATION TO DOCTOR'S LOCATION

The routing functionality of the `flutter_osm_plugin` is used to paint a map and draw a path from an origin to a destination. The map is placed at the starting location; the view is modified such that it covers both locations by zooming in on a bounding box encasing them. The `drawRoad` method enables drawing a route on the map by specifying the type of road, for example, automobile, along with visual features like width and color. This feature enables users to see a unique and interactive route between the two given locations. It also supports interactions with markers and location changes, thus enhancing user engagement and navigation.

14) BOOKING APPOINTMENTS

This feature is a crucial tool in healthcare application. This allows appointments to be booked in a very easy manner based on the availability of the nearby doctors either through call or by selecting a slot. This helps the patients in reducing their waiting time and unnecessary delay caused during the manual process. Also many elderly people will be beneficial by this feature in booking their appointment with the right doctor at the right time. Both the user and doctors have the facility to view the appointments history, ensuring better management of healthcare schedule.

15) LOCATE NEARBY PHARMACY

The "Find Pharmacy" feature utilizes the `flutter_osm_plugin` to display a map of the user's current location and nearby pharmacies as markers. The map is centered on the user's location, and pharmacies are displayed as red markers in a 1 kilometer range. The map zooms to a bounding box around the starting point, and a green marker represents the user's location. Users can interact with markers symbolizing pharmacies to view details covered in the pharmacy Details map. The details will appear in a dialog where they have options to either close the dialog or to be returned to the interface of the map. Interactive features such as this are thus capable of increasing users' efficiency in identifying and fetching information about pharmacies around them.

16) ALLEVIA-THE MEDICAL CHATBOT

Allevia is the chatbot feature in our app, designed to help users with their health-related questions. Users can type their queries, and the chatbot will provide clear and accurate answers. What makes Allevia even more user-friendly is that

TABLE 1. Comparison of alleviate and existing apps.

Feature / Parameter	Alleviate (Proposed System)	Ada Health	Babylon Health	MyFitnessPal
Primary Functionality	AI-driven personalized healthcare platform integrating symptom prediction, specialist recommendation, and prescription management.	Symptom checker using AI chatbot for self-diagnosis.	Virtual consultations and symptom assessment using AI.	Diet and fitness tracking platform.
Appointment Booking Efficiency	Real-time booking with verified doctors using IMR-based authentication.	No integrated booking; limited to symptom analysis.	Offers online consultations but limited regional access.	Does not support appointment booking.
Secure Record Sharing	End-to-end encrypted data sharing with doctors using QR-based access control for verified users.	Limited to app-level privacy; no QR-based verification.	Offers general privacy protection but not centralized verification.	No medical record sharing feature.
Prescription Management	Allows doctors to issue digital prescriptions and auto-generates QR-based summaries.	Not supported.	Offers consultation summaries only.	Not applicable.
Specialist Recommendation	Uses Logistic Regression and NLP-based analysis to suggest the most relevant medical specialist.	Suggests general practitioners based on symptoms.	Suggests specialists only after manual selection.	Not applicable.
Diet and Fitness Recommendations	Personalized diet and fitness plans generated using predictive models.	Limited dietary advice.	Focused on consultations, not lifestyle management.	Core functionality based on calorie tracking.

it allows follow-up questions, making it easy to get more details or clarification during the conversation. The chatbot runs on the Groq API, which uses a pre-trained model to understand and answer a wide range of health topics.

C. RESULTS

1) DOCTOR REGISTRATION AND VALIDATION

The doctor can register the clinic on the application in a userfriendly interface. Doctor has to enter some basic information like registration number, specialist and state medical council. The authenticity of the doctor is cross-verified based on their registration number in the Indian Medical Registry (IMR) Database using selenium based web-automation and webscraping technique. Fig. 2 indicates registration and login screen for users, Fig. 3 shows validation of doctors via IMR database, Fig. 4 shows details of a doctor after verification process.

The chatbot is designed to enhance user engagement by offering conversational support within the application. It adapts to user inputs to provide relevant information in a simple and understandable manner. By supporting interactive dialogue, the system helps users explore their queries more effectively. This feature contributes to a more responsive and user-centric healthcare experience within the application.

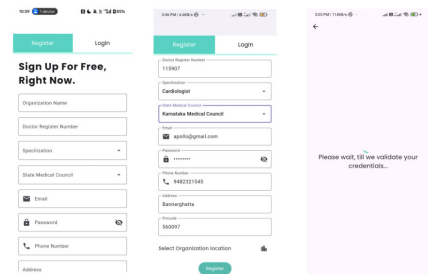


FIGURE 2. Doctor registration form.

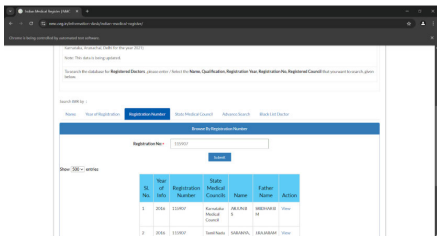


FIGURE 3. Doctor Validation via IMR.

2) SECURING PATIENT'S ELECTRONIC HEALTH RECORD (EHR) VIA QR

The doctor can quickly access the history of a patient’s health, previous illnesses, and their latest health metrics. This can

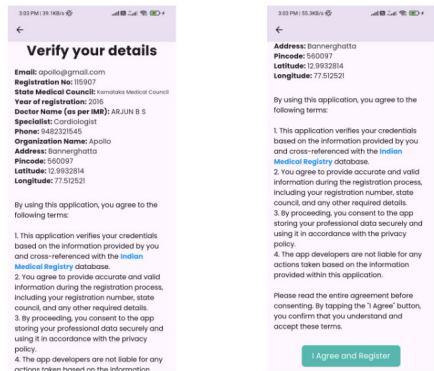


FIGURE 4. Verified doctor details.

be achieved with a quick scan of the QR code displayed on the patient’s profile page. This feature mainly aims to secure the patient’s health record by restricting access only to the authenticated doctors on the Alleviate platform, ensuring that sensitive patient’s information are kept confidential from outside world other than the registered medical professionals. This feature enables the doctors to gain deeper insights on the patients’ medical history for a more accurate evaluation, minimizing the patient’s engagement time, and helps treating more patients by increased efficiency. The code allows doctors to get patient history. Fig. 5, Fig. 6.

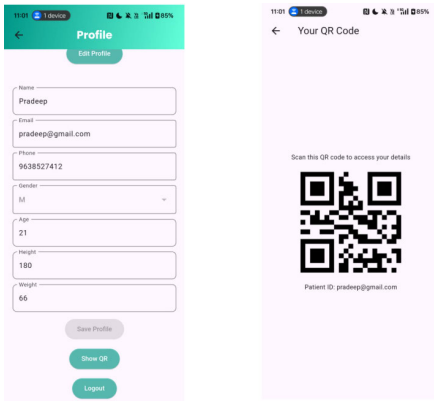


FIGURE 5. User Profile and QR Code.

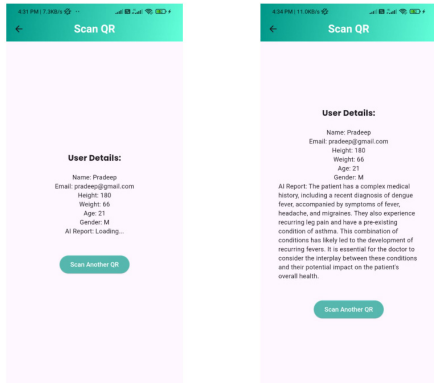


FIGURE 6. Summarized patient’s health record.

3) PRESCRIPTION MANAGEMENT

The user can manage all the prescriptions provided by doctors through the application. This facilitates easy management of all the medicines for the patient. The doctors can quickly enter the prescription on the application with an intuitive interface. The system automatically generates the PDF of the prescription report and sends it to the patient instantly. This feature ensures that individual prescriptions can be created under the registered user, providing prescription management easy for every family member. This creates a centralized hub to store the prescriptions, eliminating the need of physical copies. Fig. 7, Fig. 8.

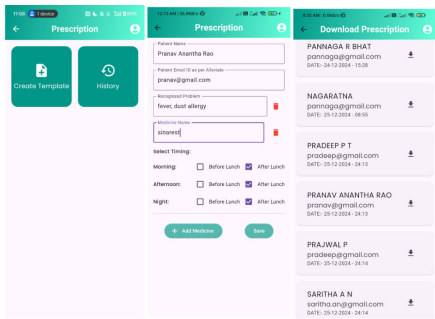


FIGURE 7. Prescription generation with logs.

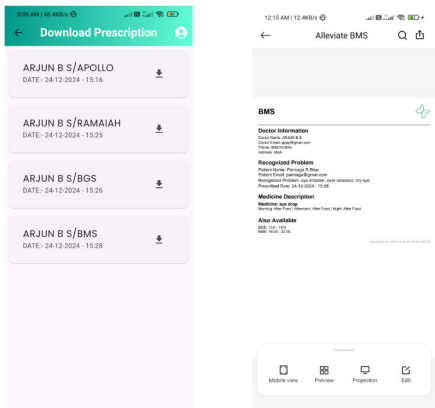


FIGURE 8. Downloadable user’s prescription history.

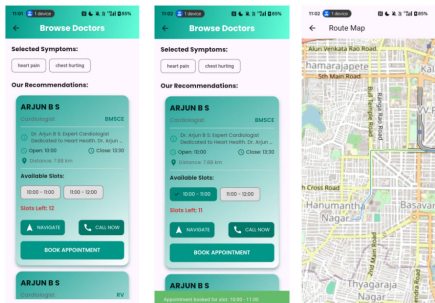


FIGURE 9. Appointments booking and navigation.

4) APPOINTMENTS BOOKING AND NAVIGATING

Patients can pre-book appointment slots through the application. The doctor can also customize the timings when the users are allowed to book appointments and set their availability. Users can book an appointment with the click of a button. Fig. 9, Fig. 10.

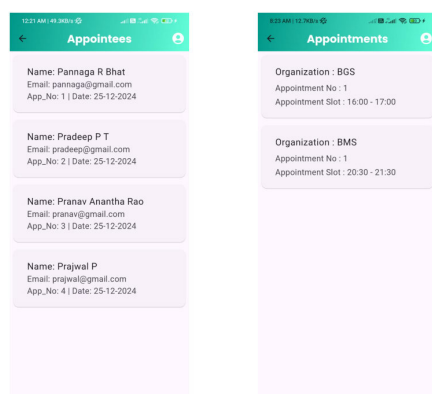


FIGURE 10. Confirmed appointments dashboard.

III. CONCLUSION

This is a development example of a healthcare application called “Alleviate,” which uses modern technologies to make both accessibility and efficiency of medical services accessible. The application includes the following functionalities: voice recognition helps users find physicians by their symptoms, find nearby pharmacies, arrange appointments, provide diet plans and exercises, and make it possible for both the users and physicians to check prescriptions. For doctors, it provides the facility of adding clinics, monitoring appointment availability, and viewing patient reports through QR codes. Altogether, these features intend to bridge the gap between patients and healthcare providers, creating an efficient and accessible healthcare environment.

The application introduces various key enhancements with respect to existing systems in various fields such as:

- **Innovative Aspects:** Voice recognition technology in conjunction with QR code scanning has significantly improved both usability and accessibility, by securing patient’s electronic health record (EHR) via QR Code.
- **Appointment Management:** Patients can receive recommendations for nearby specialists based on their symptoms. This helps in developing seamless communication between Doctors and patients, reducing barriers in appointment management and prescription access.
- **Holistic Approach:** By integrating diet plans, exercise tips, and nearby pharmacy locators, the app helps with general wellness beyond regular medical consultations.

There are some aspects that need improvement and future improvements that could significantly enhance the functionality and usability of the application:

- **Individualized Perspectives:** The current diet and exercise recommendations are generalized, but AI-based personal plans with a user’s medical history and individual preferences may greatly enhance the effectiveness and relevance of the system.
- **Scalability:** Existing infrastructure may not efficiently sustain a large number of users and wide disease predictions, but cloud solutions offer the advantage of scalability combined with high availability, therefore potentially allowing the application to increase in size without causing decreased performance.
- **Security and Privacy:** The current system does not include advanced encryption for sensitive user data, but implementing comprehensive end-to-end encryption protocols and complying with regulations such as HIPAA would significantly improve data security and user privacy.
- **Differently-abled and Elderly Persons:** More access will be made available to users of all ages and abilities as future applications are released by virtue of the accessibility features that include large font buttons, screen reader support, and voice navigation assistance for the visually or motor-disabled user.
- **Healthcare Facility Integration:** Established formal partnerships or integration with hospitals, certified clinics, and other facilities would assist with verification of doctors’ information and prescriptions and the validation of the users’ medical records.
- **RFID integration:** This next mile could benefit for greater enhancements in patient identification and checking access to medical records and retrieving medication with RFID technology [20], say, in the clinic environment.

Although the application has not yet been deployed publicly, internal testing using simulated user data has been carried out to evaluate its stability, accuracy, and responsiveness. The testing included validating the specialist prediction module, ensuring that the machine learning model returned relevant results for diverse symptom inputs, and verifying that the appointment scheduling system correctly synchronized with the backend in real time. Similarly, the prescription generation and download modules were tested across multiple devices to confirm consistency in formatting and accessibility. During testing, minor limitations such as delays in map rendering using OpenStreetMap services and occasional synchronization lags when multiple users accessed the same database were observed. The application currently supports only one language. Therefore, addressing these shortcomings and including all the proposed enhancements, Alleviate can become all-inclusive and critical software for healthcare management, by acting as a precedent for applications in this field. Besides giving it an extended reach, the impact of the improvement will be on patient outcome as well as the effectiveness of service.

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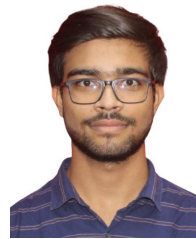


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